

The fixed shell target has a unique tracial state

MathlibAnnex.CStarAlgebra.CAR.existsUnique_tracial_state

theorem

Every tracial state is forced back to the CAR trace and hence to its unique extension.

Statement

Let A be the completed CAR algebra and τ its normalized trace. Fix a representative shell family and its chosen unitary links L_j in the selected atomic representation π . Write $T = C^*(\pi(A), \{L_j\})$ for the resulting concrete target and let $\iota : A \rightarrow T$, $\iota(a) = \pi(a)$, for its faithful unital source map. This fixes one target and one source map throughout. There exists exactly one state $\phi : T \rightarrow \mathbb{C}$ such that $\phi(ab) = \phi(ba)$ for all $a, b \in T$.

Assumptions

The representative shell family is fixed. A competing functional is required to be a state and tracial on T ; equality with τ on the source is derived, not assumed.

Conclusion

The unique tracial state is the chosen extension `traceExtension` family. This uniqueness quantifies all tracial states on T , whereas extension uniqueness quantifies all states with the prescribed source restriction.

Faithfulness of a representation and faithfulness of this state are different assertions. This theorem asserts existence and uniqueness of a tracial state. The separate result

`traceExtension_star_mul_self_eq_zero_iff` concerns state faithfulness and is not an additional hypothesis here.

Proof route

Produce the tracial extension, restrict any other tracial state to the source, and invoke the two uniqueness results in that order.

▼ Proof steps

1. The extension theorem gives a state ϕ with $\phi \circ \iota = \tau$, and `traceExtension_mul_comm` proves it tracial on T . This supplies the existence part.
2. For any tracial state ψ on T , the functional $f = \psi \circ \iota$ is continuous and complex linear. Since ι preserves 1 and products,

$$f(1) = 1, \quad f(ab) = \psi(\iota(a)\iota(b)) = \psi(\iota(b)\iota(a)) = f(ba).$$

CAR trace uniqueness, `eq_trace_of_apply_one_of_mul_comm`, therefore gives $f = \tau$. That source theorem determines values on matrix units and extends from finite stages by density; no target-separability hypothesis is used.

- Now ψ is a state extension of τ . The extension-equality theorem applies and gives $\psi = \phi$. In the source, these two steps are `apply_shellFamilySourceHom_eq_trace_of_apply_one_of_mul_comm` followed by `eq_traceExtension_of_mem_stateSpace_of_mul_comm`.

Main citations

- The stated existence or structural result —
`MathlibAnnex.CStarAlgebra.CAR.existsUnique_tracial_state`
- The fixed concrete target and source map —
`MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom`
- Faithfulness of the source embedding —
`MathlibAnnex.CStarAlgebra.CAR.shellFamilySourceHom_injective`
- The chosen extension is tracial — `MathlibAnnex.CStarAlgebra.CAR.traceExtension_mul_comm`
- Uniqueness with prescribed source trace —
`MathlibAnnex.CStarAlgebra.CAR.existsUnique_state_extension_trace`
- Uniqueness of the normalized continuous CAR trace —
`MathlibAnnex.CStarAlgebra.CAR.eq_trace_of_apply_one_of_mul_comm`
- Restriction of any target trace is the CAR trace —
`MathlibAnnex.CStarAlgebra.CAR.apply_shellFamilySourceHom_eq_trace_of_apply_one_of_mul_comm`
- Identification of an arbitrary tracial state —
`MathlibAnnex.CStarAlgebra.CAR.eq_traceExtension_of_mem_stateSpace_of_mul_comm`
- Separate state-faithfulness assertion —
`MathlibAnnex.CStarAlgebra.CAR.traceExtension_star_mul_self_eq_zero_iff`

Supporting route explanation

Lean source signature (exact)

```
theorem existsUnique_tracial_state (family : RepresentativeShellFamily) :
  ∃! ϕ : ShellFamilyTarget family → L[ℂ] ℂ,
    ϕ ∈ MathlibAnnex.Analysis.CStarAlgebra.stateSpace (ShellFamilyTarget family) ∧
    ∀ a b, ϕ (a * b) = ϕ (b * a)
```

In the source	Mathematical meaning
<code>family : RepresentativeShellFamily;</code> <code>ShellFamilyTarget family</code>	The fixed CAR shell family and once-chosen links L_j determine the same concrete target $T = C^*(\pi(A), \{L_j\})$ throughout.
<code>shellFamilySourceHom family; trace</code>	The faithful unital source map $\iota : A \rightarrow T$, $\iota(a) = \pi(a)$, and the normalized CAR trace $\tau : A \rightarrow \mathbb{C}$.

In the source	Mathematical meaning
$\exists! \phi : \text{ShellFamilyTarget family} \rightarrow \mathbb{L}[\mathbb{C}] \mathbb{C}$	There is exactly one continuous complex-linear functional $\phi : T \rightarrow \mathbb{C}$ satisfying the following state and trace conditions simultaneously.
$\phi \in \dots \text{stateSpace (ShellFamilyTarget family)}$	ϕ is positive and normalized: $\phi(1_T) = 1$.
$\forall a \ b, \ \phi(a * b) = \phi(b * a)$	For every pair $a, b \in T$, $\phi(ab) = \phi(ba)$. The quantifier includes all target elements. No source-restriction condition is imposed in this predicate; $\phi \circ \iota = \tau$ is derived in the proof before extension uniqueness is applied.

Further source notes: In contrast to the extension-uniqueness signature, it does not prescribe source values. The proof derives those values using CAR trace uniqueness before applying extension uniqueness.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.CStarAlgebra.CAR.existsUnique_tracial_state

Accepted content SHA-256: d5f53c2203ea9706a87a450b8a5bf899e58fdcadcf555dea645e6613ebc82d4

Accepted source guide SHA-256: 76bfc9157448d43db1de5128c1d282c39747e29c2f09b046e566eace27b79d8

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73